#In this code, we demonstrate the existence of the curves γ2 (r2=0) and B̃=0 in region A7 by Lemma 6.13.

$$A := -(w+z)^3\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right):$$

$$B := x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3:$$

$$R := B^2\cdot\left((w+z)^2-y^2\right)^3-A^2\cdot\left(y^2\cdot z\cdot w\right)^3$$

$$R := \left(x^2y^5w^2z^2(w+z)^2(w-z)-x^2y^2(w+z)^3\left(w^2+z^2\right)w^2z^2(w-z)+w^4y^3z^4(w+z)^3 + w^4z^4x^3y^3\right)^2\left((w+z)^2-y^2\right)^3-(w+z)^6\left(\left(x^2y^2+z^4\right)w^4-x^2y^2z^4\right)^2y^6z^3w^3 \quad \textbf{(1)}$$

#Apply the rational parametrization constructed in this section.

$$parametrization := \left\{ \mathrm{w}=\mathrm{a}+\frac{1}{a},\ z=b+\frac{1}{b},\ x=a-\frac{1}{a}+b-\frac{1}{b},\ y=a-\frac{1}{a}-b+\frac{1}{b}\right\}:$$

simplify(collect(numer(expand(subs(parametrization, R))), a));

$$-(a+b)^6(a-b)^6\Big(\left(-4b^{18}+8b^{16}-4b^{14}\right)a^{30}+\left(6b^{20}-32b^{18}+44b^{16}-32b^{14}+6b^{12}\right)a^{28}+\left(-1024b^{22}+72b^{20}+692b^{18}+8b^{16}-76b^{14}+72b^{12}\right)a^{26}+\left(-254b^{28}+840b^{26}+6504b^{24}-696b^{22}-25752b^{20}-169584b^{18}-181788b^{16}-85104b^{14}-50328b^{12}-4792b^{10}+360b^8+72b^6+2b^4\right)a^{20}+\left(b^{32}+8b^{30}+20b^{28}+1672b^{26}+1950b^{24}-32376b^{22}-157780b^{20}-317080b^{18}-279672b^{16}-222104b^{14}-125524b^{12}-7288b^{10}-3426b^8-120b^6+20b^4+8b^2+1\right)a^{16}+\left(-9b^{24}-72b^{22}+144b^{20}-1912b^{18}+142b^{16}+4232b^{14}+2192b^{12}-1864b^{10}-265b^8\right)a^8+\left(72b^{20}-76b^{18}+1288b^{16}+692b^{14}-696b^{12}-512b^{10}\right)a^6+\left(6b^{20}-32b^{18}+44b^{16}-288b^{14}\right. \quad \textbf{(2)}$$

$$- 250\, b^{12}\big)\, a^4 + \big(-4\, b^{18} + 8\, b^{16} - 4\, b^{14}\big)\, a^2 + a^{32}\, b^{16} + b^{16} + 2560 \Big(b^6 - \frac{6}{5}\, b^4 + \frac{8}{5}\, b^2$$

$$- 1\Big)\, b^7 \left(b^2+1\right)^4 a^{11} + 512\, b^7 \left(b^2-2\right) \left(b^2+1\right)^5 a^9 + 512\, b^7 \left(b^{10} - 10\, b^8 - 9\, b^6 + 42\, b^4$$

$$- 19\, b^2 + 1\big) \left(b^2+1\right)^4 a^{15} - 2560\, b^7 \Big(b^8 - \frac{16}{5}\, b^6 + b^4 - \frac{3}{5}\, b^2 + \frac{3}{5}\Big) \left(b^2+1\right)^4 a^{13}$$

$$+ 512\, b^7 \left(b^{10} + 11\, b^8 - 78\, b^6 + 81\, b^4 - 22\, b^2 + 1\right) \left(b^2+1\right)^4 a^{17} - 2560\, b^{11} \Big(b^6 - \frac{38}{5}\, b^4$$

$$+ \frac{42}{5}\, b^2 - \frac{11}{5}\Big) \left(b^2+1\right)^4 a^{21} - 1536\, b^9 \Big(b^8 - \frac{43}{3}\, b^6 + \frac{95}{3}\, b^4 - \frac{52}{3}\, b^2 + \frac{7}{3}\Big) \left(b^2$$

$$+ 1\right)^4 a^{19} - 1024 \left(b^2-5\right) \Big(b^2 - \frac{1}{2}\Big)\, b^{13} \left(b^2+1\right)^4 a^{23} - 4\, b^2 \left(b^{28} + 12\, b^{26} + 37\, b^{24}\right.$$

$$+ 378\, b^{22} + 4187\, b^{20} + 19656\, b^{18} + 41514\, b^{16} + 57358\, b^{14} + 65514\, b^{12} + 34120\, b^{10}$$

$$\left. + 4315\, b^8 + 1018\, b^6 - 155\, b^4 + 12\, b^2 + 1\right) a^{14} - 4\, b^2 \left(b^{28} + 76\, b^{26} - 539\, b^{24} - 2118\, b^{22}\right.$$

$$+ 5211\, b^{20} + 36360\, b^{18} + 92394\, b^{16} + 78414\, b^{14} + 38058\, b^{12} + 17992\, b^{10} - 2085\, b^8$$

$$\left. - 518\, b^6 + 37\, b^4 + 12\, b^2 + 1\right) a^{18} + 8\, b^6 \Big(b^{20} + b^{18} - 699\, b^{16} + 2369\, b^{14} - \frac{1037}{2}\, b^{12}$$

$$- 1827\, b^{10} + \frac{2163}{2}\, b^8 - 159\, b^6 - 27\, b^4 + b^2 + 1\Big) a^{22} - 1024\, b^9 \left(b^2+1\right)^4 a^7 + 8 \Big(b^{20}$$

$$+ b^{18} - 27\, b^{16} + 33\, b^{14} - \frac{2189}{2}\, b^{12} - 3875\, b^{10} - \frac{9357}{2}\, b^8 - 3231\, b^6 - 2747\, b^4 - 1055\, b^2$$

$$- 127\Big)\, b^6\, a^{10} + 2\, b^4 \left(b^{24} + 36\, b^{22} + 180\, b^{20} - 1372\, b^{18} - 9036\, b^{16} - 25400\, b^{14} - 54798\, b^{12}\right.$$

$$\left. - 64184\, b^{10} - 27212\, b^8 - 6620\, b^6 - 2636\, b^4 + 36\, b^2 + 1\right) a^{12} - 9\, b^8 \Big(b^{16} + \frac{4168}{9}\, b^{14}$$

$$- 528\, b^{12} - 328\, b^{10} + \frac{3698}{9}\, b^8 - 72\, b^6 - 16\, b^4 + 8\, b^2 + 1\Big) a^{24}\Big) \left(a^2+1\right)^3 (a\, b$$

$$+ 1)^{12} \left(b^2+1\right)^3$$

#Define r2

$$r2 := \Big(b^{16} + a^{32}\, b^{16} - 1024\, b^9 \left(b^2+1\right)^4 a^7 + 8 \Big(b^{20} + b^{18} - 27\, b^{16} + 33\, b^{14} - \frac{2189}{2}\, b^{12}$$

$$- 3875\, b^{10} - \frac{9357}{2}\, b^8 - 3231\, b^6 - 2747\, b^4 - 1055\, b^2 - 127\Big)\, b^6\, a^{10} + 2\, b^4 \left(b^{24} + 36\, b^{22}\right.$$

$$+ 180\, b^{20} - 1372\, b^{18} - 9036\, b^{16} - 25400\, b^{14} - 54798\, b^{12} - 64184\, b^{10} - 27212\, b^8 - 6620\, b^6$$

$$- 2636\, b^4 + 36\, b^2 + 1\big)\, a^{12} - 4\, b^2\, \big(b^{28} + 12\, b^{26} + 37\, b^{24} + 378\, b^{22} + 4187\, b^{20} + 19656\, b^{18}$$

$$+ 41514\, b^{16} + 57358\, b^{14} + 65514\, b^{12} + 34120\, b^{10} + 4315\, b^8 + 1018\, b^6 - 155\, b^4 + 12\, b^2 + 1\big)$$

$$a^{14} - 4\, b^2\, \big(b^{28} + 76\, b^{26} - 539\, b^{24} - 2118\, b^{22} + 5211\, b^{20} + 36360\, b^{18} + 92394\, b^{16} + 78414\, b^{14}$$

$$+ 38058\, b^{12} + 17992\, b^{10} - 2085\, b^8 - 518\, b^6 + 37\, b^4 + 12\, b^2 + 1\big)\, a^{18} + 8\, \Big(b^{20} + b^{18} - 699\, b^{16}$$

$$+ 2369\, b^{14} - \frac{1037}{2}\, b^{12} - 1827\, b^{10} + \frac{2163}{2}\, b^8 - 159\, b^6 - 27\, b^4 + b^2 + 1\Big)\, b^6\, a^{22} - 9\, \Big(b^{16}$$

$$+ \frac{4168}{9}\, b^{14} - 528\, b^{12} - 328\, b^{10} + \frac{3698}{9}\, b^8 - 72\, b^6 - 16\, b^4 + 8\, b^2 + 1\Big)\, b^8\, a^{24} + \big(-4\, b^{18}$$

$$+ 8\, b^{16} - 4\, b^{14}\big)\, a^{30} + \big(6\, b^{20} - 32\, b^{18} + 44\, b^{16} - 32\, b^{14} + 6\, b^{12}\big)\, a^{28} + \big(-1024\, b^{22} + 72\, b^{20}$$

$$+ 692\, b^{18} + 8\, b^{16} - 76\, b^{14} + 72\, b^{12}\big)\, a^{26} + \big(-254\, b^{28} + 840\, b^{26} + 6504\, b^{24} - 696\, b^{22}$$

$$- 25752\, b^{20} - 169584\, b^{18} - 181788\, b^{16} - 85104\, b^{14} - 50328\, b^{12} - 4792\, b^{10} + 360\, b^8 + 72\, b^6$$

$$+ 2\, b^4\big)\, a^{20} + \big(b^{32} + 8\, b^{30} + 20\, b^{28} + 1672\, b^{26} + 1950\, b^{24} - 32376\, b^{22} - 157780\, b^{20}$$

$$- 317080\, b^{18} - 279672\, b^{16} - 222104\, b^{14} - 125524\, b^{12} - 7288\, b^{10} - 3426\, b^8 - 120\, b^6 + 20\, b^4$$

$$+ 8\, b^2 + 1\big)\, a^{16} + \big(-9\, b^{24} - 72\, b^{22} + 144\, b^{20} - 1912\, b^{18} + 142\, b^{16} + 4232\, b^{14} + 2192\, b^{12}$$

$$- 1864\, b^{10} - 265\, b^8\big)\, a^8 + \big(72\, b^{20} - 76\, b^{18} + 1288\, b^{16} + 692\, b^{14} - 696\, b^{12} - 512\, b^{10}\big)\, a^6$$

$$+ \big(6\, b^{20} - 32\, b^{18} + 44\, b^{16} - 288\, b^{14} - 250\, b^{12}\big)\, a^4 + \big(-4\, b^{18} + 8\, b^{16} - 4\, b^{14}\big)\, a^2 - 1024\, \big(b^2$$

$$- 5\big)\, \Big(b^2 - \frac{1}{2}\Big)\, \big(b^2 + 1\big)^4\, b^{13}\, a^{23} + 2560\, \Big(b^6 - \frac{6}{5}\, b^4 + \frac{8}{5}\, b^2 - 1\Big)\, \big(b^2 + 1\big)^4\, b^7\, a^{11}$$

$$+ 512\, b^7\, \big(b^2 - 2\big)\, \big(b^2 + 1\big)^5\, a^9 + 512\, b^7\, \big(b^{10} - 10\, b^8 - 9\, b^6 + 42\, b^4 - 19\, b^2 + 1\big)\, \big(b^2$$

$$+ 1\big)^4\, a^{15} - 2560\, \big(b^2 + 1\big)^4\, \Big(b^8 - \frac{16}{5}\, b^6 + b^4 - \frac{3}{5}\, b^2 + \frac{3}{5}\Big)\, b^7\, a^{13} + 512\, b^7\, \big(b^{10} + 11\, b^8$$

$$- 78\, b^6 + 81\, b^4 - 22\, b^2 + 1\big)\, \big(b^2 + 1\big)^4\, a^{17} - 2560\, \Big(b^6 - \frac{38}{5}\, b^4 + \frac{42}{5}\, b^2 - \frac{11}{5}\Big)\, \big(b^2$$

$$+ 1\big)^4\, b^{11}\, a^{21} - 1536\, \Big(b^8 - \frac{43}{3}\, b^6 + \frac{95}{3}\, b^4 - \frac{52}{3}\, b^2 + \frac{7}{3}\Big)\, \big(b^2 + 1\big)^4\, b^9\, a^{19}\Big) :$$

#Define $\tilde{B}$

$$\tilde{B} := \Big(\Big(-\frac{1}{2}\, b^5 + b^7\Big)\, a^{10} + \Big(\frac{3}{2}\, b^7 - 3\, b^5 + b^3\Big)\, a^8 + \frac{b^2\, \big(b^2 - 5\big)\, \big(b^2 + 1\big)^3\, a^7}{2} + \Big(-\frac{1}{2}\, b$$

$$- \frac{7}{2}\, b^5 + 3\, b^3\Big)\, a^6 + \Big(\frac{1}{2} - \frac{3}{2}\, b^8 - 4\, b^6 - 3\, b^4\Big)\, a^5 + \Big(\frac{5}{2}\, b^3 - b - \frac{1}{2}\, b^7\Big)\, a^4 - \big(b^2$$

$$+1)^3 a^3 + \left(b^5 - \frac{1}{2} b\right) a^2 - \frac{b^3}{2}\Bigg) :$$

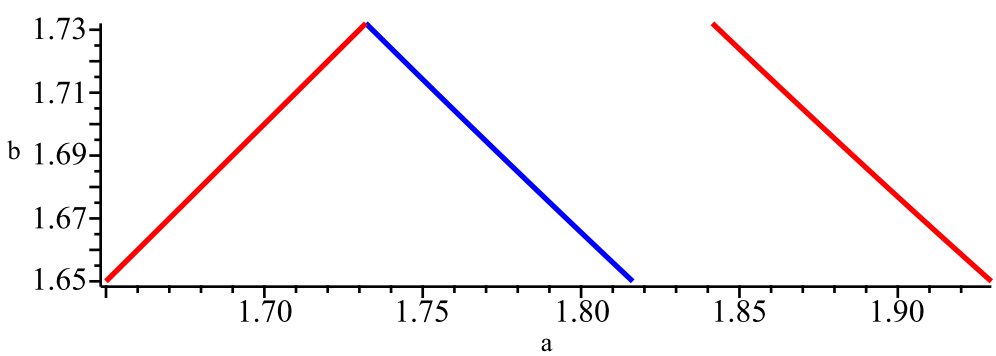
#Plot the curves ñ=0 and p̃=0 in region A7.

$r := a - b :$

with(plots) :

```
implicitplot( [r2 = 0, B̃ = 0, r = 0], a = 1 .. 1.95, b = 1.65 .. sqrt(3),
        scaling = constrained,
        gridrefine = 2,
        thickness = 2,
        color = [blue, red, red],
        labels = ["a", "b"],
         title = "Graphic for  r2=0 and B̃=0 in A7");
```

Graphic for r2=0 and B̃=0 in A7



#Apply the Möbius transformation on the rectangle A7 = [sqrt(3), 1.95) × [1.65, sqrt(3)) *constructing r2*

$$moebius := \left\{ a = \frac{\mathrm{sqrt}(3) + \left(\frac{39}{20}\right) c}{1 + c}, \right.$$

$$b = \frac{\frac{33}{20} + (\mathrm{sqrt}(3))\, d}{1 + d} \Bigg\} :$$

$$\hat{r}2 := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(\mathrm{moebius}, \mathrm{r2}))), c) :$$

#Compute the degree of r2 in c

$degree(r2, c);$

$$32 \tag{3}$$

#Define the coefficients ci = coeff(r2, c, i)

$seq(evalf(assign(cat('c', i), coeff(r2, c, i))), i = 0 .. 32);$

#Compute the all degree of ci in d

$seq(degree(eval(cat('c', i)), d), i = 0 .. 32);$

$$31, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32, 32 \tag{4}$$

#From here we will analyze the signal variation of all coefficients ci

$seq(sign(evalf(coeff(c32, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{5}$$

$seq(sign(evalf(coeff(c31, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{6}$$

$seq(sign(evalf(coeff(c30, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{7}$$

$seq(sign(evalf(coeff(c29, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{8}$$

$seq(sign(evalf(coeff(c28, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{9}$$

$seq(sign(evalf(coeff(c27, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{10}$$

$seq(sign(evalf(coeff(c26, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{11}$$

$seq(sign(evalf(coeff(c25, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{12}$$

$seq(sign(evalf(coeff(c24, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{13}$$

$seq(sign(evalf(coeff(c23, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{14}$$

$seq(sign(evalf(coeff(c22, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{15}$$

$seq(sign(evalf(coeff(c21, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{16}$$

$seq(sign(evalf(coeff(c20, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (17)$$

$seq(sign(evalf(coeff(c19, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (18)$$

$seq(sign(evalf(coeff(c18, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (19)$$

$seq(sign(evalf(coeff(c17, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (20)$$

$seq(sign(evalf(coeff(c16, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (21)$$

$seq(sign(evalf(coeff(c15, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (22)$$

$seq(sign(evalf(coeff(c14, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (23)$$

$seq(sign(evalf(coeff(c13, d, i))), i = 32 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \quad (24)$$

$seq(sign(evalf(coeff(c12, d, i))), i = 32 .. 0, -1);$
$fsolve(c12 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1$$

$$0.06100868716 \quad (25)$$

$seq(sign(evalf(coeff(c11, d, i))), i = 32 .. 0, -1);$
$fsolve(c11 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1$$

$$0.1568562538 \quad (26)$$

$seq(sign(evalf(coeff(c10, d, i))), i = 32 .. 0, -1);$
$fsolve(c10 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1$$

$$0.2718844641 \quad (27)$$

$seq(sign(evalf(coeff(c9, d, i))), i = 32 .. 0, -1);$
$fsolve(c9 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$0.4124868395 \quad (28)$$

$seq(sign(evalf(coeff(c8, d, i))), i = 32 .. 0, -1);$
$fsolve(c8 = 0, d, 0 .. \text{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$0.5882536070 \quad (29)$$

$seq(sign(evalf(coeff(c7, d, i))), i = 32 .. 0, -1);$

$fsolve(c7 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$0.8142550629 \tag{30}$$

$seq(sign(evalf(coeff(c6, d, i))), i = 32 .. 0, -1);$
$fsolve(c6 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$1.115608298 \tag{31}$$

$seq(sign(evalf(coeff(c5, d, i))), i = 32 .. 0, -1);$
$fsolve(c5 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$1.537523994 \tag{32}$$

$seq(sign(evalf(coeff(c4, d, i))), i = 32 .. 0, -1);$
$fsolve(c4 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$2.170423416 \tag{33}$$

$seq(sign(evalf(coeff(c3, d, i))), i = 32 .. 0, -1);$
$fsolve(c3 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$3.225289385 \tag{34}$$

$seq(sign(evalf(coeff(c2, d, i))), i = 32 .. 0, -1);$
$fsolve(c2 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$5.335070180 \tag{35}$$

$seq(sign(evalf(coeff(c1, d, i))), i = 32 .. 0, -1);$
$fsolve(c1 = 0, d, 0 .. \text{infinity});$

$1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$11.66450682 \tag{36}$$

$seq(sign(evalf(coeff(c0, d, i))), i = 32 .. 0, -1);$
$coeff(c0, d, 32)$

$1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1$

$$0 \tag{37}$$

#Apply the Möbius transformation on the rectangle A7 = [sqrt(3), 1.95) × [1.65, sqrt(3)) *constructing* $\hat{B}$

$\hat{B} := \mathrm{collect}(\mathrm{numer}(\mathrm{expand}(\mathrm{subs}(moebius, \tilde{B}))), d) :$

#Compute the degree of $\hat{B}$ *in c*

$degree(\hat{B}, c);$

$$10 \tag{38}$$

#Define the coefficients Bi = coeff$(\hat{B}, c, i)$

$seq(evalf(assign(cat('B', i), coeff(\hat{B}, c, i))), i = 0 .. 10);$

#Compute the all degree of Bi in d

$seq(degree(eval(cat('B', i)), d), i = 0 .. 10);$

$$10, 10, 10, 10, 10, 10, 10, 10, 10, 10, 10 \tag{39}$$

#From here we will analyze the signal variation of all coefficients Bi

$seq(sign(evalf(coeff(B10, d, i))), i = 10 .. 0, -1);$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \tag{40}$$

$seq(sign(evalf(coeff(B9, d, i))), i = 10 .. 0, -1);$

$fsolve(B9 = 0, d, 0 .. \mathrm{infinity});$

$$1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1$$

$$0.03776075486 \tag{41}$$

$seq(sign(evalf(coeff(B8, d, i))), i = 10 .. 0, -1);$

$fsolve(B8 = 0, d, 0 .. \mathrm{infinity});$

$$1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1$$

$$0.4183483000 \tag{42}$$

$seq(sign(evalf(coeff(B7, d, i))), i = 10 .. 0, -1);$

$fsolve(B7 = 0, d, 0 .. \mathrm{infinity});$

$$1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1$$

$$1.238253787 \tag{43}$$

$seq(sign(evalf(coeff(B6, d, i))), i = 10 .. 0, -1);$

$fsolve(B6 = 0, d, 0 .. \mathrm{infinity});$

$$1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$4.288938703 \tag{44}$$

$seq(sign(evalf(coeff(B5, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{45}$$

$seq(sign(evalf(coeff(B4, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{46}$$

$seq(sign(evalf(coeff(B3, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(47)}$$

$seq(sign(evalf(coeff(B2, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(48)}$$

$seq(sign(evalf(coeff(B1, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(49)}$$

$seq(sign(evalf(coeff(B0, d, i))), i = 10 .. 0, -1);$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \quad \textbf{(50)}$$